Input/Output

- **Unix Philosophy:** "Everything is a file." Whether it's a disk file, a keyboard, or a network socket, you interact with it using the same set of system calls.
- **Windows Philosophy:** Everything is an Object managed by the Object Manager. I/O is handled via Handles to these objects, often using the I/O Request Packet (IRP) model.

Unix Files

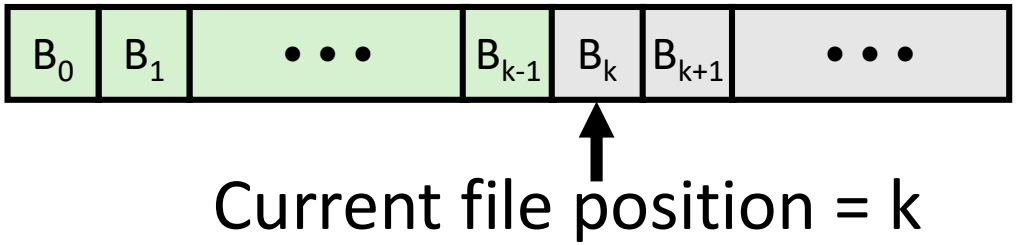
- **A Unix *file* is a sequence of *m* bytes:**
  - $B_0, B_1, \ldots, B_k, \ldots, B_{m-1}$
- **All I/O devices are represented as files:**
  - **`/dev/sda2`** (**`/usr`** disk partition)
  - **`/dev/tty2`** (terminal)
- **Even the kernel is represented as a file:**
  - **`/dev/kmem`** (kernel memory image)
  - **`/proc`** (kernel data structures)

Unix File Types

- **Regular file**
  - File containing user/app data (binary, text, whatever)
  - OS does not know anything about the format
    - other than “sequence of bytes”, akin to main memory
- **Directory file**
  - A file that contains the names and locations of other files
- **Character special and block special files**
  - Terminals (character special) and disks (block special)
- **FIFO (named pipe)**
  - A file type used for inter-process communication
- **Socket**
  - A file type used for network communication between processes

Unix I/O

- **Key Features**
  - Elegant mapping of files to devices allows kernel to export simple interface called Unix I/O
  - Important idea: All input and output is handled in a consistent and uniform way
- **Basic Unix I/O operations (system calls):**
  - Opening and closing files
    - **`open()`** and **`close()`**
  - Reading and writing a file
    - **`read()`** and **`write()`**
  - Changing the ***current file position*** (seek)
    - indicates next offset into file to read or write
    - **`lseek()`**



Core System Calls: Open, Read, Write, Close

Feature	Unix System Call	Windows API (Win32)
Open	open(path, flags, mode)	CreateFile(...) (used for both creation and opening)
Close	close(fd)	CloseHandle(hObject)
Read	read(fd, buf, count)	ReadFile(hFile, lpBuffer, ...)
Write	write(fd, buf, count)	WriteFile(hFile, lpBuffer, ...)

- **Short Counts.** In Unix, read and write might return fewer bytes than requested (a short count). This isn't an error; it happens due to:
  - Encountering **EOF** (End of File).
  - Reading from a terminal or network socket.
  - Interruption by a signal.
  - Windows handles this similarly but often relies on explicit "bytes transferred" pointers in the API call.

Opening Files

- **Opening a file informs the kernel that you are getting ready to access that file**

```
int fd;   /* file descriptor */

if ((fd = open("/etc/hosts", O_RDONLY)) < 0) {
   perror("open");
   exit(1);
}
```

- **Returns a small identifying integer *file descriptor***
  - **`fd == -1`** indicates that an error occurred
- **Each process created by a Unix shell begins life with three open files associated with a terminal:**
  - 0: standard input
  - 1: standard output
  - 2: standard error

Closing Files

- Closing a file informs the kernel that you are finished accessing that file

```
int fd;     /* file descriptor */
int retval; /* return value */

if ((retval = close(fd)) < 0) {
   perror("close");
   exit(1);
}
```

- Closing an already closed file is a recipe for disaster in threaded programs (more on this later)
- Moral: Always check return codes, even for seemingly benign functions such as `close()`

Reading Files

- **Reading a file copies bytes from the current file position to memory, and then updates file position**

```
char buf[512];
int fd;        /* file descriptor */
int nbytes;    /* number of bytes read */

/* Open file fd ...  */
/* Then read up to 512 bytes from file fd */
if ((nbytes = read(fd, buf, sizeof(buf))) < 0) {
   perror("read");
   exit(1);
}
```

- **Returns number of bytes read from file `fd` into `buf`**
  - Return type **`ssize_t`** is signed integer
  - **`nbytes < 0`** indicates that an error occurred
  - ***Short counts*** (**`nbytes < sizeof(buf)`**) are possible and are not errors!

Writing Files

- **Writing a file copies bytes from memory to the current file position, and then updates current file position**

```
char buf[512];
int fd;       /* file descriptor */
int nbytes;   /* number of bytes read */

/* Open the file fd ... */
/* Then write up to 512 bytes from buf to file fd */
if ((nbytes = write(fd, buf, sizeof(buf)) < 0) {
   perror("write");
   exit(1);
}
```

- **Returns number of bytes written from `buf` to file `fd`**
  - **`nbytes < 0`** indicates that an error occurred
  - As with reads, short counts are possible and are not errors!

Metadata and Representation

- **File Metadata**
  - **Unix:** Uses the `stat` structure. You access it via `stat()`, `fstat()`, or `lstat()`. It includes the **inode** number, size, permissions, and timestamps.
  - **Windows:** Uses `GetFileInformationByHandle`. Metadata includes file attributes (Hidden, System, Archive), which are more complex than Unix's simple bitmask permissions.
- **How the Kernel Represents Open Files**
  - **Descriptor Table (Unix):** Each process has its own table of pointers to the open file table.
  - **Open File Table:** A system-wide table containing the current file position (offset), reference count, and a pointer to the v-node/i-node.
  - **v-node/i-node Table:** Contains the actual file metadata and location on disk.
  - **Windows** uses a similar "**Handle Table**" for each process, which points to executive file objects in kernel space.

File Metadata

- ***Metadata*** **is data about data, in this case file data**
- **Per-file metadata maintained by kernel**
  - accessed by users with the **`stat`** and **`fstat`** functions

```
/* Metadata returned by the stat and fstat functions */
struct stat {
    dev_t         st_dev;      /* device */
    ino_t         st_ino;      /* inode */
    mode_t        st_mode;     /* protection and file type */
    nlink_t       st_nlink;    /* number of hard links */
    uid_t         st_uid;      /* user ID of owner */
    gid_t         st_gid;      /* group ID of owner */
    dev_t         st_rdev;     /* device type (if inode device) */
    off_t         st_size;     /* total size, in bytes */
    unsigned long st_blksize;  /* blocksize for filesystem I/O */
    unsigned long st_blocks;   /* number of blocks allocated */
    time_t        st_atime;    /* time of last access */
    time_t        st_mtime;    /* time of last modification */
    time_t        st_ctime;    /* time of last change */
};
```

Example of Accessing File Metadata

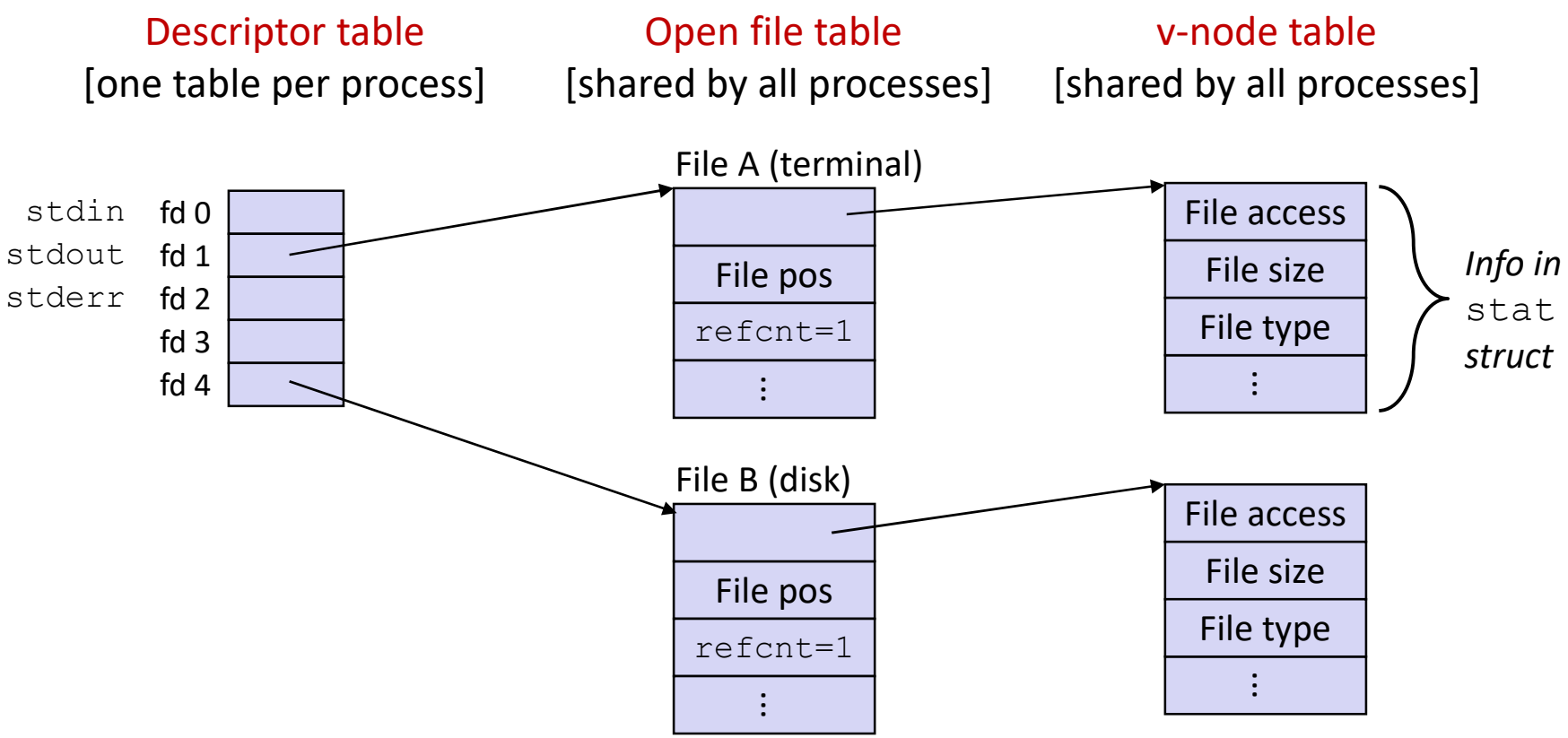
```
/* statcheck.c - Querying and manipulating a file's meta data */
#include "csapp.h"

int main (int argc, char **argv)
{
    struct stat stat;
    char *type, *readok;

    Stat(argv[1], &stat);
    if (S_ISREG(stat.st_mode))
        type = "regular";
    else if (S_ISDIR(stat.st_mode))
        type = "directory";
    else
        type = "other";
    if ((stat.st_mode & S_IRUSR)) /* OK to read?*/
        readok = "yes";
    else
        readok = "no";

    printf("type: %s, read: %s\n", type, readok);
    exit(0);
}
```

statcheck.c

```
unix> ./statcheck statcheck.c
type: regular, read: yes
unix> chmod 000 statcheck.c
unix> ./statcheck statcheck.c
type: regular, read: no
unix> ./statcheck ..
type: directory, read: yes
unix> ./statcheck /dev/kmem
type: other, read: yes
```

How the Unix Kernel Represents Open Files

- **Two descriptors referencing two distinct open disk files. Descriptor 1 (stdout) points to terminal, and descriptor 4 points to open disk file**

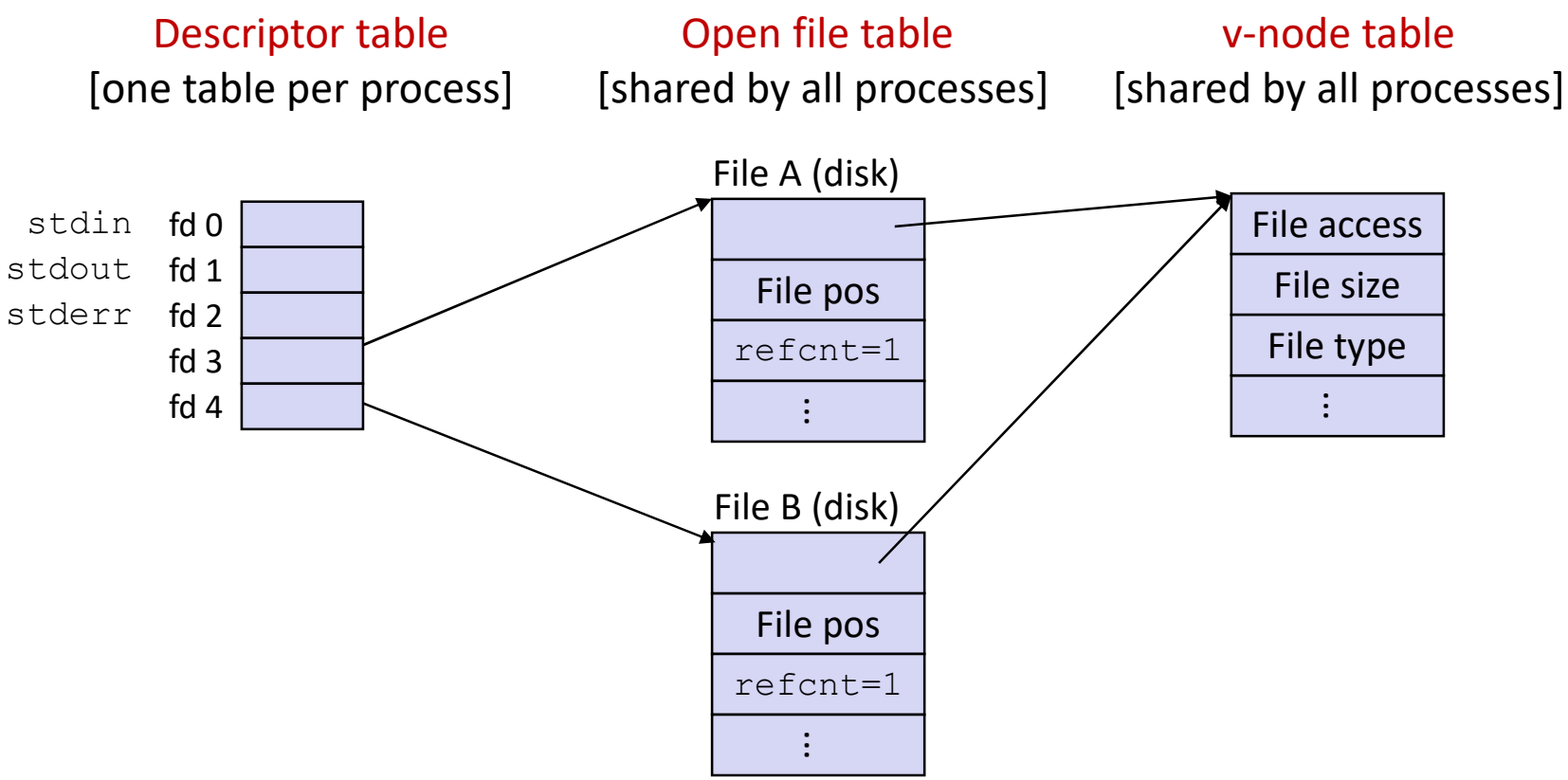


File Sharing and Redirection

- **File Sharing**
  - **Unix:** Multiple processes can share an entry in the open file table (sharing the file offset) if a process is `fork()`-ed.
  - **Windows:** Sharing is explicitly managed by "Share Modes" (`FILE_SHARE_READ`, `FILE_SHARE_WRITE`) during the `CreateFile` call.
- **I/O Redirection**
  - **Unix:** Uses the `dup2(oldfd, newfd)` system call to copy file descriptors.
  - **Mechanism:** To redirect `stdout` to a file, the shell closes FD 1 and `dup2`s the file's FD into slot 1.
  - **Windows:** Accomplished via `SetStdHandle`.

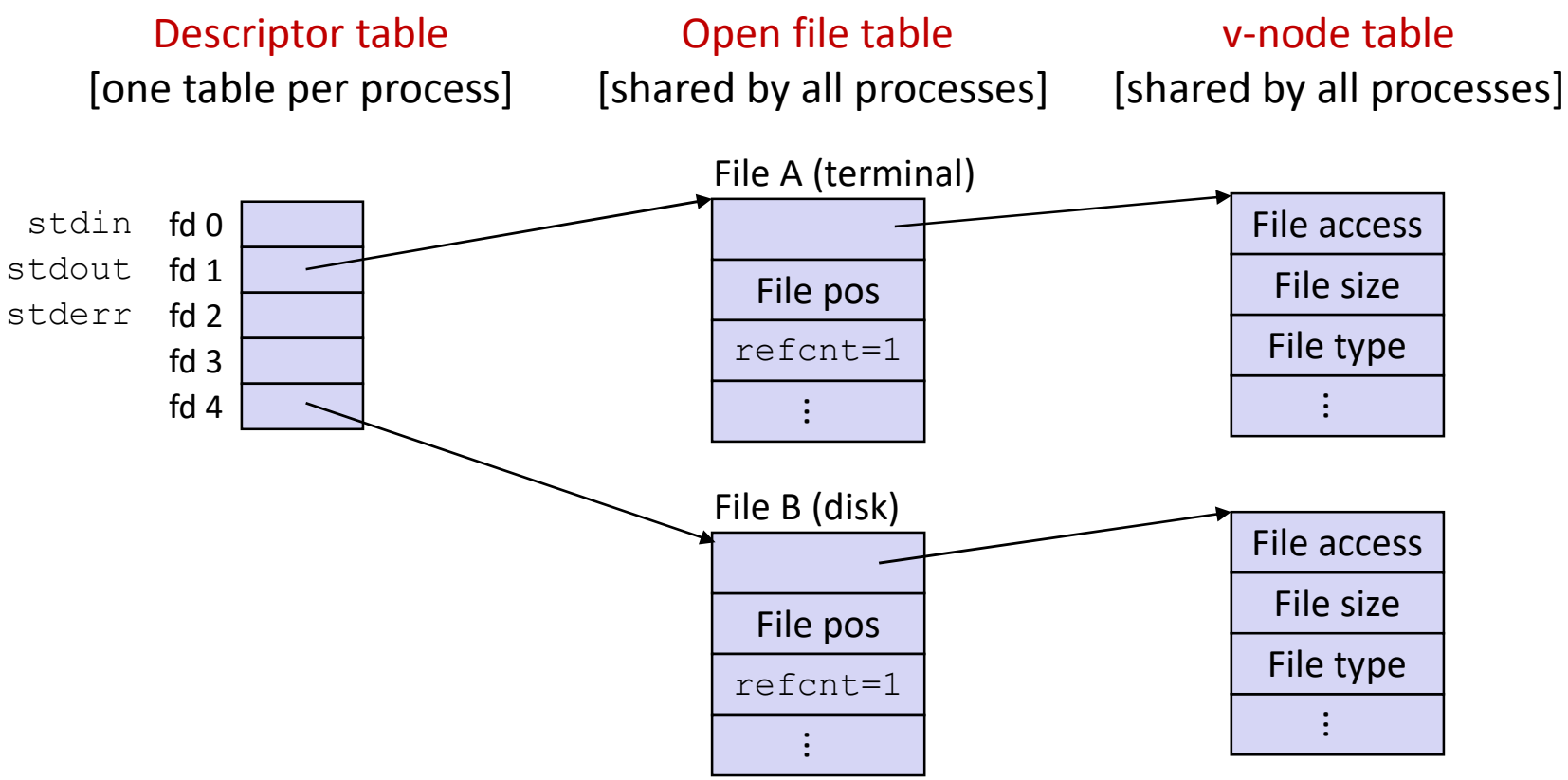
File Sharing

- **Two distinct descriptors sharing the same disk file through two distinct open file table entries**
  - E.g., Calling **`open`** twice with the same **`filename`** argument



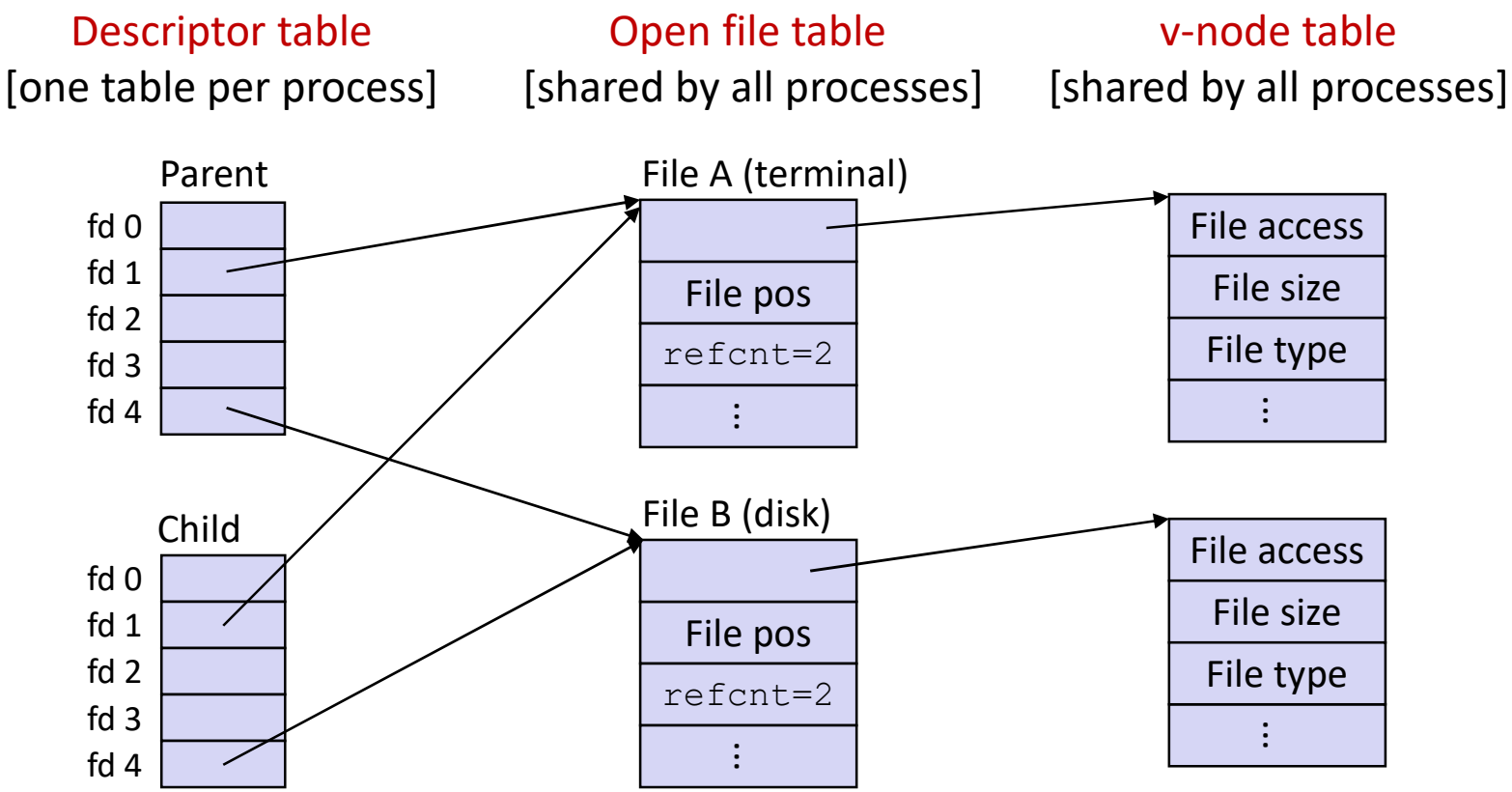
How Processes Share Files: Fork()

- **A child process inherits its parent's open files**
  - Note: situation unchanged by **`exec`** functions (use **`fcntl`** to change)
- ***Before*** **fork() call:**



How Processes Share Files: Fork()

- **A child process inherits its parent's open files**
- ***After* fork():**
  - Child's table same as parent's, and +1 to each refcnt



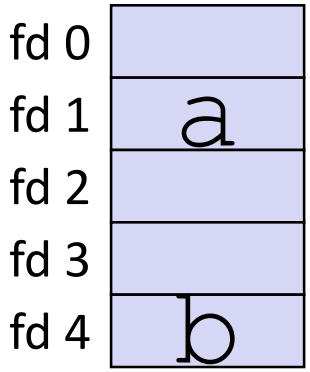
I/O Redirection

- **Question: How does a shell implement I/O redirection?**

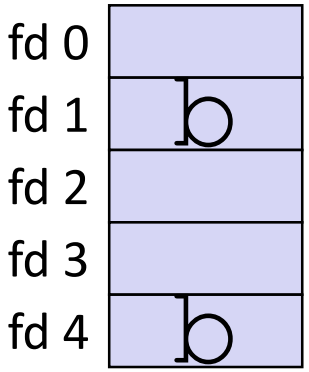
```
unix> ls > foo.txt
```

- **Answer: By calling the `dup2(oldfd, newfd)` function**
  - Copies (per-process) descriptor table entry **`oldfd`** to entry **`newfd`**

Descriptor table
before `dup2(4,1)`

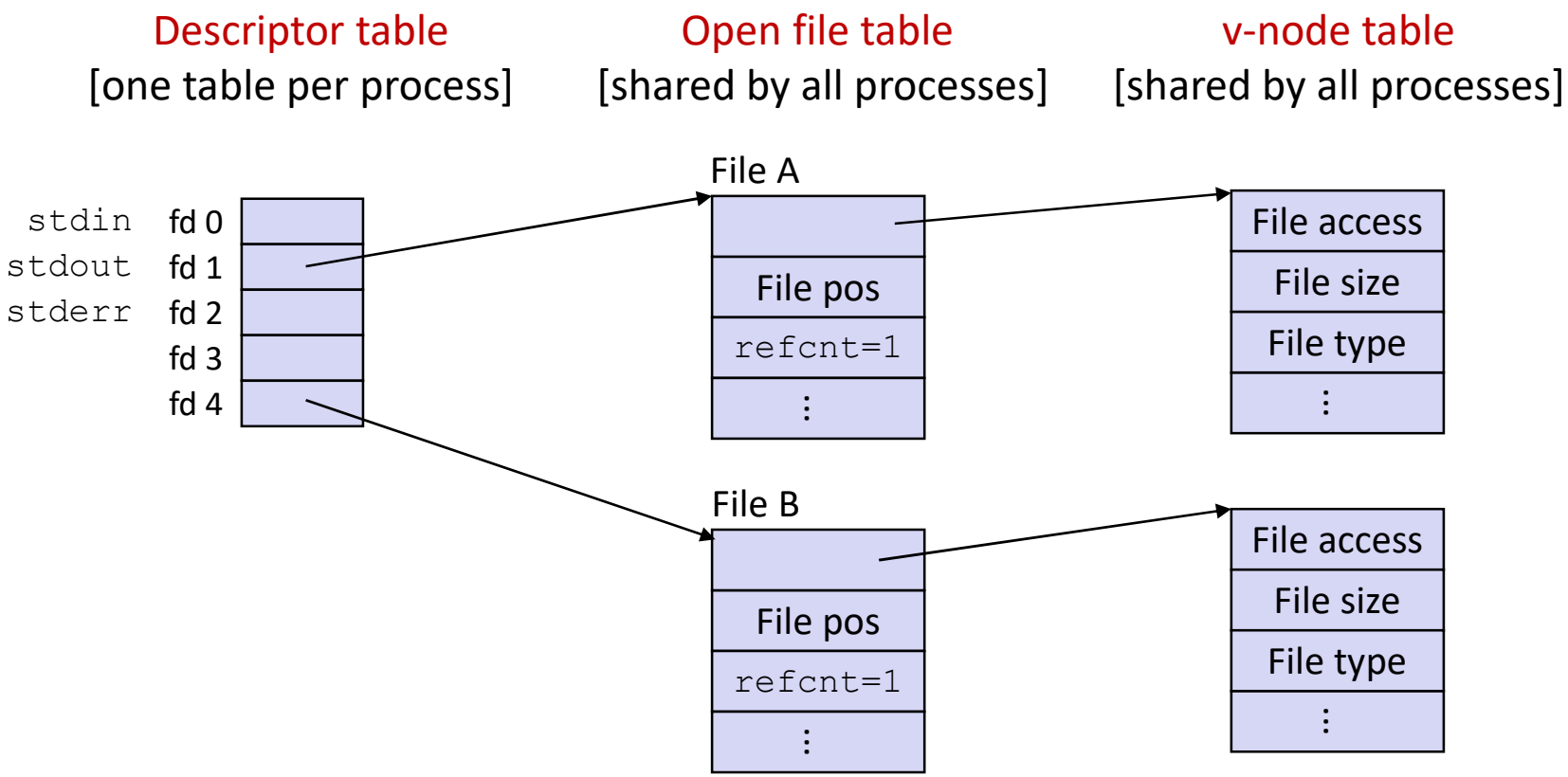


Descriptor table
after `dup2(4,1)`



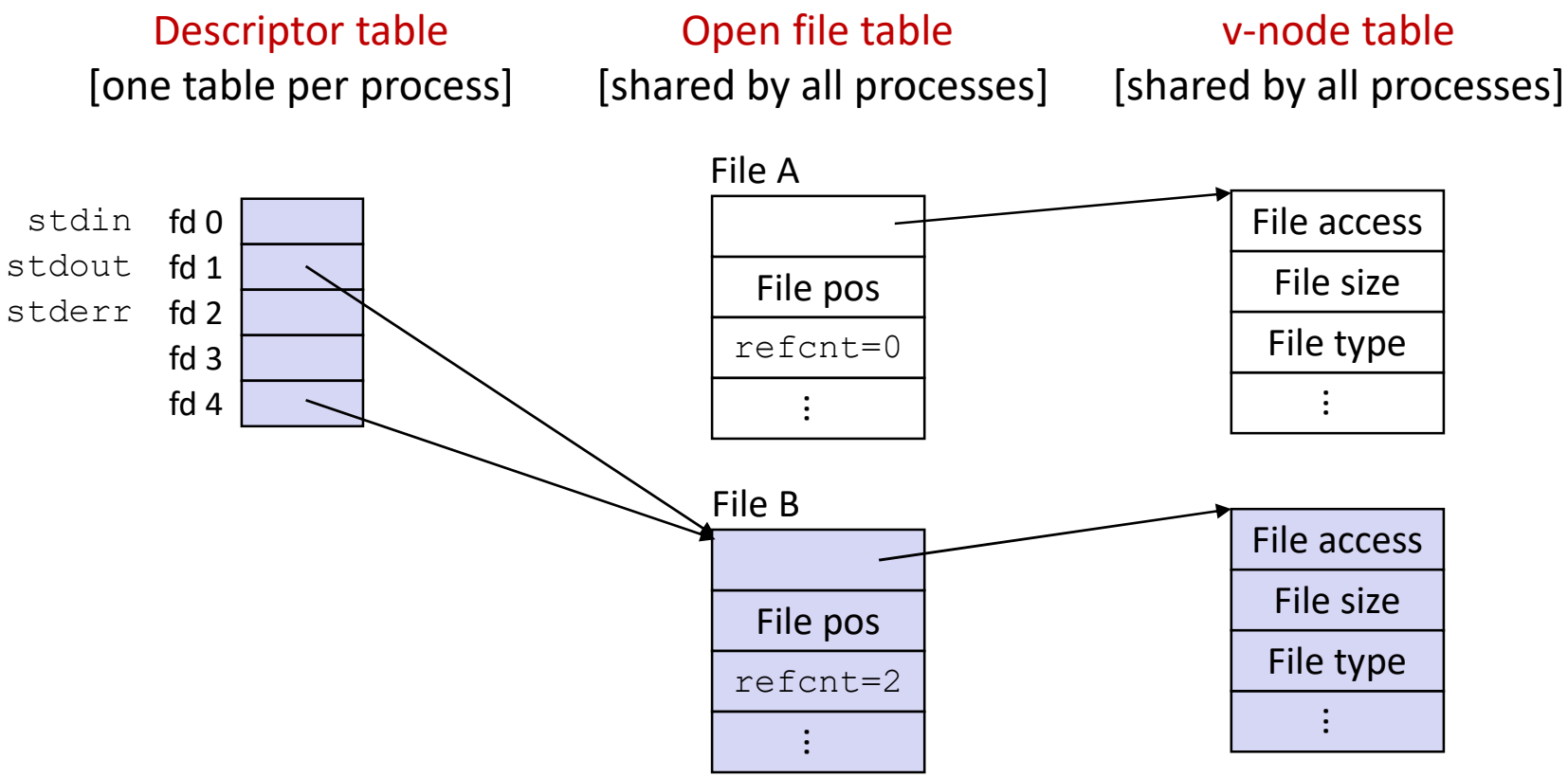
I/O Redirection Example

- **Step #1: open file to which stdout should be redirected**
  - Happens in child executing shell code, before **`exec`**



I/O Redirection Example (cont.)

- **Step #2: call `dup2(4,1)`**
  - cause fd=1 (stdout) to refer to disk file pointed at by fd=4



Standard I/O, Streams, and Buffering

While the system calls are "Unbuffered I/O," most use the Standard I/O Library (C standard library).

- **Streams:** An abstraction (`FILE *` in C) that wraps the raw file descriptor.
- **Buffered I/O:** To reduce expensive system calls, the library maintains a buffer in user memory.
  - **Full Buffering:** Writes only when the buffer is full (default for files).
  - **Line Buffering:** Writes on every newline `\n` (default for terminal/stdout).
  - **No Buffering:** Writes immediately (default for stderr).

Binary vs. Text Files

- **Unix:** No distinction. A file is a sequence of bytes.
- **Windows:** In "Text Mode," the library automatically translates `\n` (newline) to `\r\n` (carriage return + line feed) on write, and vice versa on read.

Standard I/O Functions

- **The C standard library (`libc.so`) contains a collection of higher-level *standard I/O* functions**
  - Documented in Appendix B of K&R
- **Examples of standard I/O functions:**
  - Opening and closing files (**`fopen`** and **`fclose`**)
  - Reading and writing bytes (**`fread`** and **`fwrite`**)
  - Reading and writing text lines (**`fgets`** and **`fputs`**)
  - Formatted reading and writing (**`fscanf`** and **`fprintf`**)

Standard I/O Streams

- **Standard I/O models open files as *streams***
  - Abstraction for a file descriptor and a buffer in memory
- **C programs begin life with three open streams (defined in `stdio.h`)**
  - **`stdin`** (standard input)
  - **`stdout`** (standard output)
  - **`stderr`** (standard error)

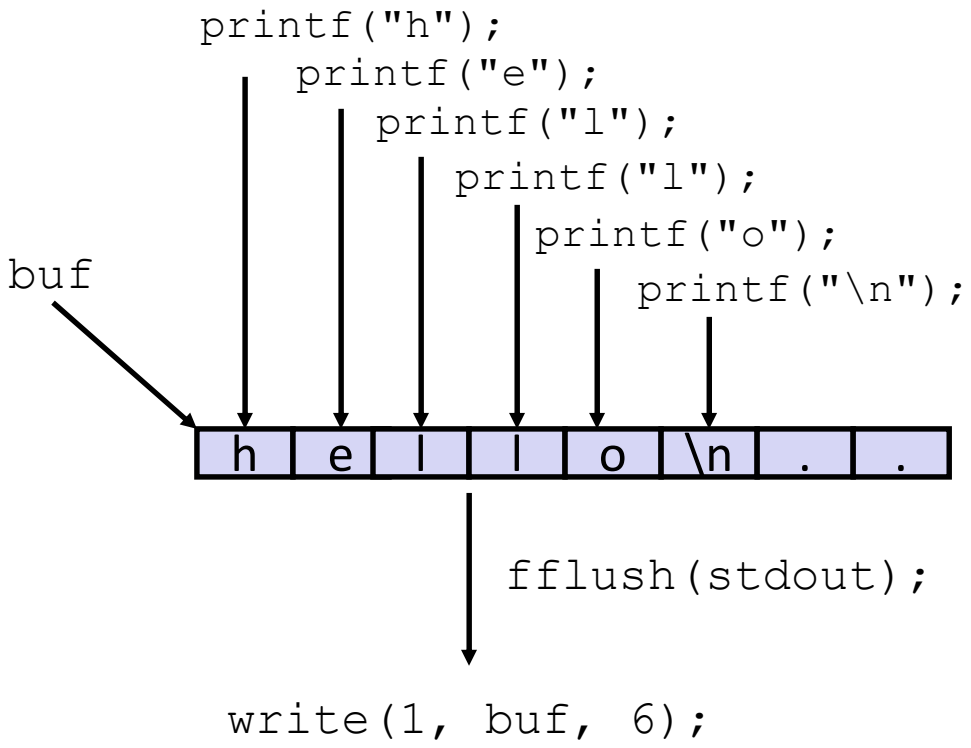
```
#include <stdio.h>
extern FILE *stdin;  /* standard input  (descriptor 0) */
extern FILE *stdout; /* standard output (descriptor 1) */
extern FILE *stderr; /* standard error  (descriptor 2) */

int main() {
    fprintf(stdout, "Hello, world\n");
}
```

Buffered I/O: Motivation

- **Applications often read/write one character at a time**
  - `getc, putc, ungetc`
  - `gets, fgets`
    - Read line of text on character at a time, stopping at newline
- **Implementing as Unix I/O calls expensive**
  - `read` and `write` require Unix kernel calls
    - > 10,000 clock cycles
- **Solution: Buffered read**
  - Use Unix `read` to grab block of bytes
  - User input functions take one byte at a time from buffer
    - Refill buffer when empty

Buffering in Standard I/O

- **Standard I/O functions use buffered I/O**



- **Buffer flushed to output fd on "\n" or `fflush()` call**

Standard I/O Buffering in Action

- You can see this buffering in action for yourself, using the always fascinating Unix `strace` program:

```
#include <stdio.h>

int main()
{
    printf("h");
    printf("e");
    printf("l");
    printf("l");
    printf("o");
    printf("\n");
    fflush(stdout);
    exit(0);
}
```

```
linux> strace ./hello
execve("./hello", ["hello"], [/* ... */]).
...
write(1, "hello\n", 6)                  = 6
...
exit_group(0)                           = ?
```

Pros and Cons of Unix I/O

- **Pros**
  - Unix I/O is the most general and lowest overhead form of I/O.
    - All other I/O packages are implemented using Unix I/O functions.
  - Unix I/O provides functions for accessing file metadata.
  - Unix I/O functions are async-signal-safe and can be used safely in signal handlers.
- **Cons**
  - Dealing with short counts is tricky and error prone.
  - Efficient reading of text lines requires some form of buffering, also tricky and error prone.
  - Both of these issues are addressed by the standard I/O and RIO packages.

Pros and Cons of Standard I/O

- **Pros:**
  - Buffering increases efficiency by decreasing the number of **`read`** and **`write`** system calls
  - Short counts are handled automatically
- **Cons:**
  - Provides no function for accessing file metadata
  - Standard I/O functions are not async-signal-safe, and not appropriate for signal handlers.
  - Standard I/O is not appropriate for input and output on network sockets
    - There are poorly documented restrictions on streams that interact badly with restrictions on sockets (CS:APP2e, Sec 10.9)

Physical I/O Organization

How Devices are Represented

- **Unix:** Special files in `/dev/`. **Character devices** (keyboards) provide a stream of bytes; **Block devices** (disks) allow random access.
- **Windows:** Devices have internal names like `\Device\HarddiskVolume1`. Users access them via symbolic links like `C:` or `COM1`.

Physical Organization

- **Interrupts:** Hardware signals the CPU when an I/O operation is complete.
- **DMA (Direct Memory Access):** Allows hardware to transfer data directly to RAM without constant CPU intervention.
- **Device Drivers:** The bridge between the OS kernel and the specific hardware registers.